STANDARD QUALITY PLAN

ITEM: Rolling Shutter

SQP No : RE / QA / SQP / C / 005
Rev. No. : 00
Date : 01.03.2019
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Approved by QA

SL NO	COMPONENT/ OPERATION	CHARACTERISTICS	CATEGORY OF CHECK	TYPE OF CHECK	QUANTUM OF CHECK		REFERENCE DOCUMENT	ACCEPTANCE NORMS	FORMAT OF RECORD		INSPECTION AGENCY			REMARKS
					M	C/O					M	C	O	
1	2	3	4	5	6		7	8	9	D*	**10			11
1.0	**Raw Material Verification.**	a. **Cold Rolled Steel** Strips used for rolling shutter lath section.	Major	TC Review	100%	100%	IS : 6248 (Latest Revision)	**NOTE 1**	TC	√	P	V	V	**Note 1:** Confirming to Temper No. 5 Dead soft quality of IS 4030 (LR). Thickness of the steel strips should not be less than 0.9 mm for shutter up to & including 3.5 m width AND not less than 1.2 mm for shutter above 3.5 m width.
		b. **Mild Steel Sections of Hot Rolled Steel** used for Guide channels , brackets & lock plate.	Major	TC Review	100%	100%	IS : 6248 (Latest Revision)	**NOTE 2**	TC	√	P	V	V	**NOTE 2:** Thickness > 3.15 mm. Free from surface defects and edges cleanly sheared. Confirming to IS 5986 (latest revision)
		c. **Steel Pipes** used for the suspension shaft of the roller.	Major	TC Review	100%	100%	IS : 6248 (Latest Revision)	**NOTE 3**	TC	√	P	V	V	**NOTE 3:** Heavy duty Pipe. Confirming to IS 1161 (latest revision)

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		d. **Cast Iron Casting** used for roller pulley wheels, U – clamps & bevel gears.	Major	TC Review	100%	100%	IS : 6248 (Latest Revision)	**NOTE 4**	TC	√	P	V	V	**NOTE 4:** Casting shall be free from blow holes, surface defects such as cracks, burns etc. Confirming to Grade 15 of IS 210 (latest revision)
		e. **Springs** used in the roller for counter balancing the rolling shutter.	Major	TC Review	100%	100%	IS : 6248 (Latest Revision)	**NOTE 5**	TC	√	P	V	V	**Note 5:** Springs made either from high tensile spring steel wire or flat spring steel strip. The spring steel wire shall Confirming to Grade 2 of IS 4454 (Part I) (latest revision) Flat spring steel strip shall be from 0.8 to 1.0 % carbon steel strip,
		f. **Malleable Cast Iron** used for clips.	Major	TC Review	100%	100%	IS : 6248 (Latest Revision)	**NOTE 6**	TC	√	P	V	V	**Note 6:** Confirm to IS 2108 (latest revision)

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1	2	3	4	5	6		7	8	9	D*	**10			11
		g. **Aluminum Alloy Extrusions** for the components of rolling shutters of aluminum.	Major	TC Review	100%	100%	IS : 6248 (Latest Revision)	**NOTE 7**	TC	√	P	V	V	**Note 7:** Confirm to 53000 (NE 5) or 64430 (HE 30) of IS 733 (latest revision)
2.0	Fabrication of various parts of rolling shutter.	a. Curtain. b. Lock Plate c. Guide Channels d. Bracket Plate. e. Roller f. Hood Covers g. Gear warms h. Fixing Bolts i. Safety Devices	Major	Physical	100%	100%	IS : 6248 (Latest Revision)	**Cl. No. 7** FABRICATION of IS:6248 (Latest Revision)	IIR	√	P	P	V	Fabrication requirement for various parts or rolling shutter laid in IS 6248 needs to be followed by vendor and shall be verified by owner during final inspection.

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					M	C/O					M	C	O	
1	2	3	4	5	6		7	8	9	D*	**10			11
3.0	Operation. (Among these three types of rolling shutter operation system, one shall be applicable based on PO requirement.)	a. **Self Coiling** Type Rolling Shutter.	Major	Physical	100%	100%	IS : 6248 (Latest Revision)	**Cl. No. 8.1** FABRICATION of IS : 6248 (Latest Revision)	IIR		P	P	V	
		b. **Gear Operated** Type Rolling Shutter.	Major	Physical	100%	100%	IS : 6248 (Latest Revision)	**Cl. No. 8.2** FABRICATION of IS : 6248 (Latest Revision)	IIR		P	P	V	
		c. **Electrically Operated** Rolling Shutter.	Major	Physical	100%	100%	IS : 6248 (Latest Revision)	**Cl. No. 8.3** FABRICATION of IS : 6248 (Latest Revision)	IIR		P	P	V	Electric motor should be operating on 400/440 V, 3 phase, 50 cycles AC supply.
4.0	Painting	a. Surface Preparation.	Major	Visual	100%	100%	IS : 6248 (Latest Revision)	**NOTE 8**	IIR		P	V	V	**NOTE 8** Surface should be free from rust, oil, grease or any other foreign material prior to start painting.

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		b. Primer Application (Red oxide **Zinc Chrome** Primer)	Major	Measure ment	100%	100%	IS : 6248 (Latest Revision)	DFT 30 to 35 micron	IIR		P	V	V	Primer & Final Paint needs to be carried out at manufacturing shop and touch up shall be carried out by vendor at site after complete installation.
		c. Finished **Synthetic Enamel** Paint Two Shop Coat	Major	Measure ment	100%	100%	IS : 6248 (Latest Revision)	DFT 30 micron for each shop coat.	IIR		P	V	V	
5.0	Dimension	Width, height & thickness of Rolling shutter with accessories.	Major	Measure ment	100%	100%	**NOTE 9**	**NOTE 9**	IIR		P	V	H	**NOTE 9** Measurement sheet of **Door Openings** jointly measured by Vendor & Owner's Site Representative.

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6.0	Marking & Packing	Marking on Rolling Shutter including it's accessories & packing list.	Major	Physical verificati on	100%	100%	IS : 6248 (Latest Revision) and Packing List	IS : 6248 (Latest Revision) and Packing List	Packing List		P	P	H	

LEGEND:-

D* Records identified with tick (√) shall be essentially included by supplier & manufacturer in Quality Documentation package.

** **M**: Manufacturer / Sub-Supplier, **C**: Main Supplier , **O**: OWNER or their authorized representative.

Use the following term as appropriate in columns 10. **P**: Perform,**V**: Verification and **H**: Customer Hold Point to be witnessed and work shall not proceeded till it is witnessed and cleared in writing.

TC: Test Certificate, **TR**: Test Report.